

Max-Cut problems in graphs

1. Randomized Large-Cut algorithm for random $G_{n,p}$ graphs

We implement the algorithm, the way it's described in our theory, that is by assigning each available vertex on the A, B sets, independently and uniformly at random and then counting the number of edges that cross the partition.

When our realization of the randomized algorithm is executed, we obtain the value of $C(A, B)$, the number of edges m , the max number of iterations $\frac{m}{2} + 1$ as well as the number of actual iterations (*samples*) made until a cut with value greater than or equal to $\frac{m}{2}$ is found.

By analyzing the results printed in the MATLAB command window, in many runs of the experiment, it is clear that the actual samples needed in order to obtain a Large-Cut, are significantly less than the upper limit of iterations defined in the exercise ($\frac{m}{2} + 1$). This means that the upper limit set on the number of samples is very loose, and the algorithm does a great job in finding $C(A, B) \geq \frac{m}{2}$ remarkably sooner than the upper limit.

Such a result is to be expected, since our theory states that the expected number of samples before finding a Large-Cut is smaller than or equal to $\frac{m}{2} + 1$.

Also, it is important to note that when it comes to counting the edges that cross the cut, our algorithm does so using two *for* loops (one in order to get each edge from the edges list E , and one nested to the first for checking the existence of its vertices in A, B , using the *existsIn()* function). This means that our implementation agrees with theory, in terms of complexity (polynomial time).

2. Deterministic Large-Cut algorithm for random $G_{n,p}$ graphs

The algorithm implemented in this part is deterministic in the sense that there is some logic applied in the placement of the vertices in A, B (greedy algorithm), as opposed to the randomized case, where each vertex is placed u.a.r .

Because of its greedy nature, this algorithm divides the vertices in a ratio stretched away from 50-50 (as it roughly happens in the randomized version). This is done because the deterministic algorithm assigns vertices to each set so that the partition created acquires the largest value possible, in each step, resulting in a an unpredictable ratio, which depends on how many edges exists in the graph and in what (arbitrary) order the graph is traversed.

It is important to state that, in both cases, the value of the cut (although it will always be larger than $\frac{m}{2}$) will heavily depend on the number of edges n , as well as the probability p of an edge to be created in the $G_{n,p}$ graph.

A combination of small p and n may result in a sparse graph, thus not many edges created giving a "small" Large-Cut.

m -Balls and n -Bins experiment

After implementing and running the experiment K times, we obtain and plot the number of the bins that are empty, in a histogram.

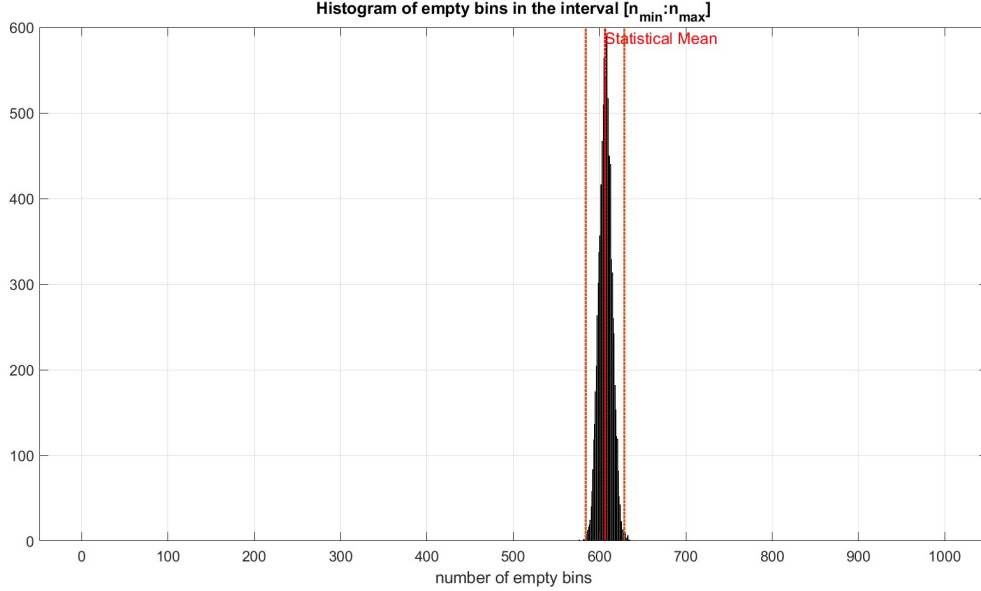


Figure 1: Number of empty bins for $K = 10^4$ iterations of the m -Balls and n -Bins experiment. The statistical mean $\mathcal{E}[F]$ is shown in red, while in orange we highlight the interval $[\mathcal{E}[F] - \epsilon, \mathcal{E}[F] + \epsilon]$ concentrating almost all of the empty bin values.

With the help of the drawn lines, we can clearly see that almost every value of the K empty bins experiments is located in an interval $[\mathcal{E}[F] - \epsilon, \mathcal{E}[F] + \epsilon]$, with $\epsilon = \sqrt{m}$. This is because, the m -Balls and n -Bins experiment is mathematically represented as a function F of all the thrown balls, which is considered to be Lipschitz with bound 1, thus giving us:

$$Pr(|F - \mathcal{E}[F]| \geq \epsilon) \leq 2e^{-\frac{2\epsilon^2}{m}}$$

In words, the above formula says that values with distance ϵ or more from the mean, are at most $2e^{-\frac{2\epsilon^2}{m}}$ likely to exceed the $[\mathcal{E}[F] - \epsilon, \mathcal{E}[F] + \epsilon]$ interval.

So by arbitrarily choosing $\epsilon = \sqrt{m}$, it is very unlikely ($p_{max} = \frac{2}{e^2} \approx 0.27$) that we report any values out of the $[\mathcal{E}[F] - \sqrt{m}, \mathcal{E}[F] + \sqrt{m}]$ interval.

This argument explains why in the histogram above, almost no value of the reported empty bins falls outside of the orange lines, denoting the two ends of the concentration interval.